PHYSICAL DESIGN

TABLE OF CONTENTS

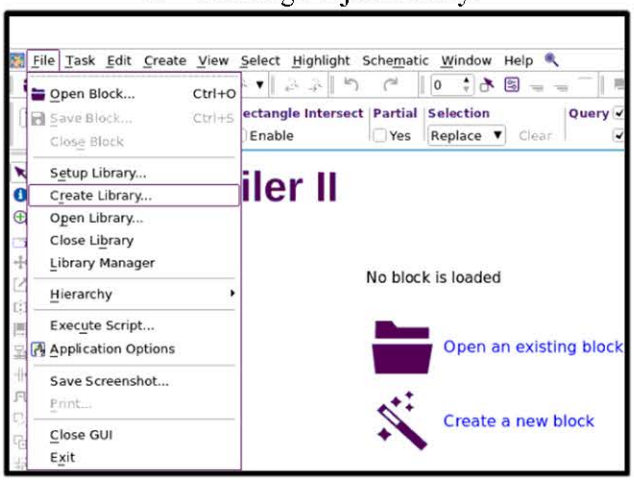
S.NO	TITLE
1	**A**PPLICATION **S**PECIFIC **I**NTEGRATED **C**IRCUIT (ASIC) FLOW
2	**P**HYSICAL **D**ESIGN (PD) FLOW
3	TOOLS USED
4	INPUTS AND OUTPUTS OF PD FLOW AND EXPLANATION
5	CREATING PROJECT FOLDER
6	INVOKING OF SHELL
7	IMPORT DESIGN
8	FLOORPLAN
9	POWERPLAN
10	PLACEMENT
11	**C**LOCK **T**REE **S**YNTHESIS (CTS)
12	ROUTING
13	SPEF FILE GENERATION
14	**S**TATIC **T**IMING **A**NALYSIS (STA)
15	**P**HYSICAL **V**ERIFICATION (PV)

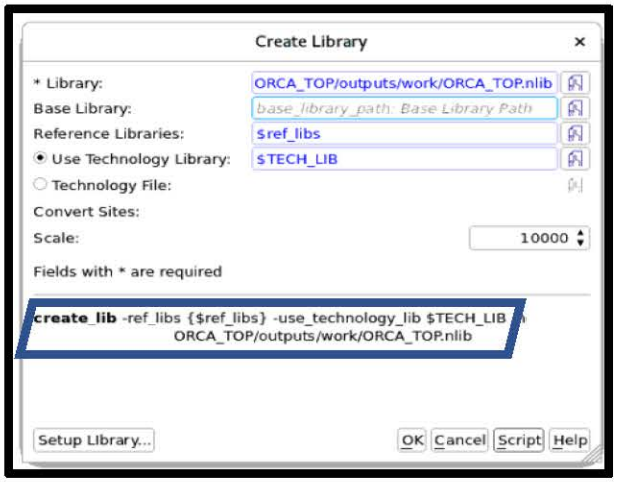
PHYSICAL DESIGN

IMPORT DESIGN

a. To open Graphical User Interface (GUI)
icc2_shell > start_gui

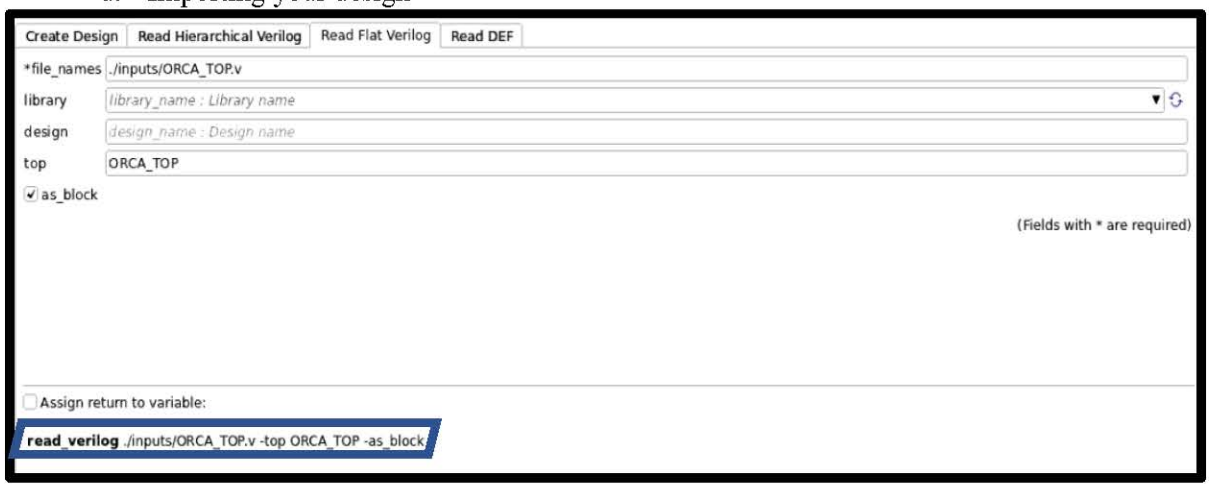
b. Creating Project library.

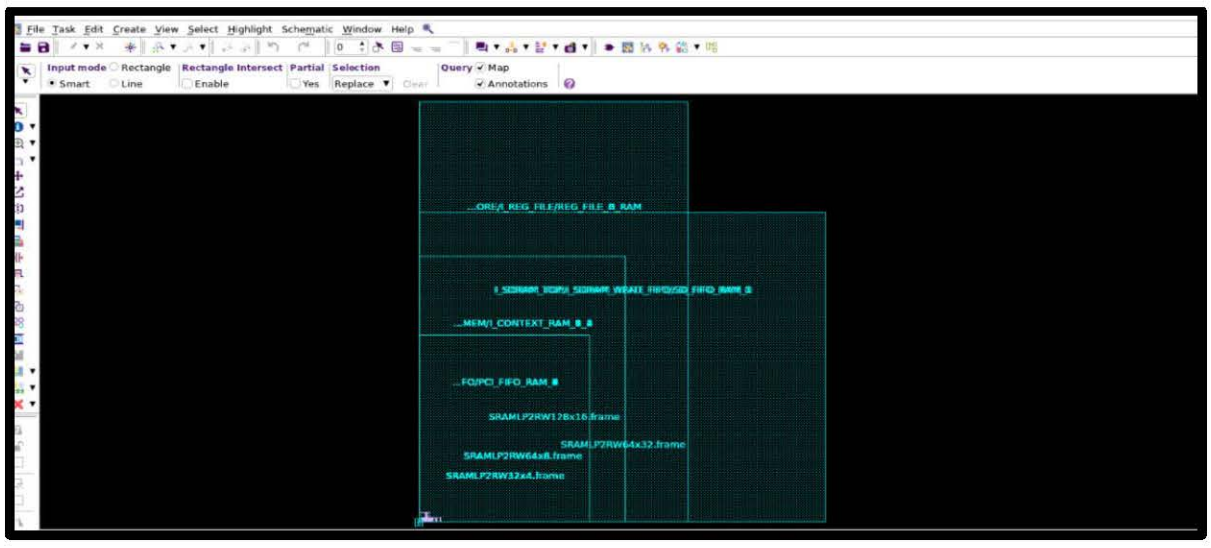




c. Setting up required library variables (i.e. target and link libraries)
icc2_shell > set_app_var link_library " saed32_1p9m_tech saed32_hvt saed32_lvt saed32_rvt saed32_sram_lp"*

d. Importing your design





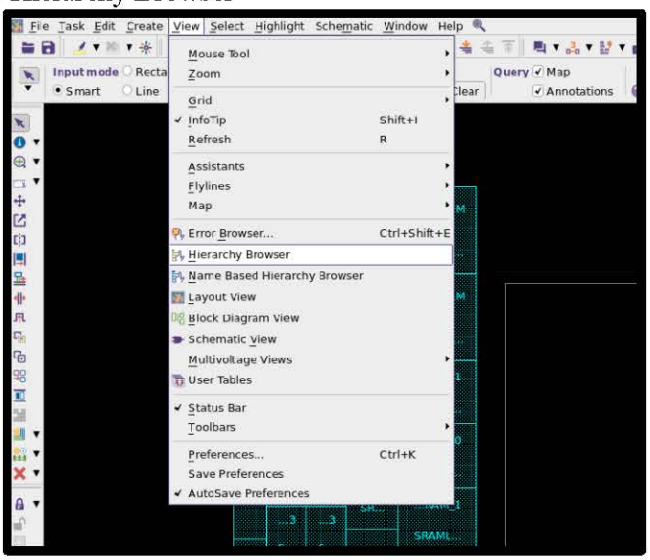
e. Reading your parasitic values (R &C)?

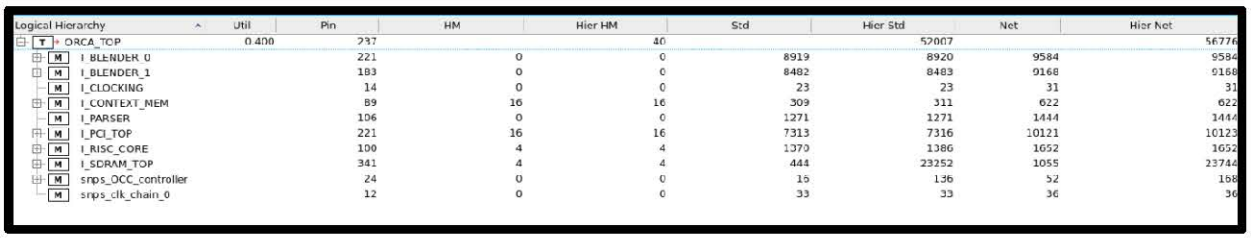
icc2_shell > read_parasitic_tech -tlup ${max_tlup} -layermap ${map} -name worst_parameter
icc2_shell > read_parasitic_tech -tlup ${min_tlup} -layermap ${map} -name best_parameter

f. To read the timing constraints

icc2_shell> source ./inputs/sdc_constraints/mcmm_ORCA_TOP.tcl

g. Showing the Hierarchy Browser





Logical Hierarchy	Util	Pin	HM	Hier HM	Std	Hier Std	Net	Hier Net
ORCA_TOP	0.400	237		40		52007		56776
I_BLENDER_0		221	0	0	8919	8920	9584	9584
I_BLENDER_1		183	0	0	8482	8483	9168	9168
I_CLOCKING		14	0	0	23	23	31	31
I_CONTEXT_MEM		89	16	16	309	311	622	622
I_PARSER		106	0	0	1271	1271	1444	1444
I_PCI_TOP		221	16	16	7313	7316	10121	10123
I_RISC_CORE		100	4	4	1370	1386	1652	1652
I_SDRAM_TOP		341	4	4	444	23252	1055	23744
snps_OCC_controller		24	0	0	15	136	52	168
snps_clk_chain_0		12	0	0	33	33	36	36

h. Sanity Checks
   i. To check whether the netlist/design is proper or not.
   ii. To check whether the libraries are correct or not
   iii. Check if mismatches are present between the current design and reference libraries

i. Save Block and Library

PHYSICAL DESIGN

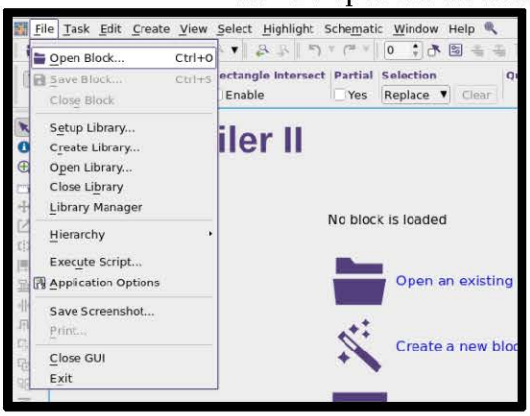
PURPOSE	ICC2 COMMAND/VARIABLE
To set LINK LIBRARIES	link_library
To create PROJECT LIBRARY	create_lib
IMPORT DESIGN	read_verilog
To read PARASATIC INFORMATION	read_parasitic_tech
To read TIMING CONSTRAINTS	read_sdc
To get the count of a) Ports b) Macros c) Standard cells	 a) sizeof_collection [get_ports *] b) sizeof_collection [get_flat_cells -filter "is_hard_macro"] c) sizeof_collection [get_flat_cells -filter "!is_hard_macro"]
SANITY CHECKS a) to check the netlist b) to check timing c) mismatch	 a) check_netlist b) check_timing c) report_design_mismatch
To save the design a) Library b) Block/CEL	 a) save_lib b) save_block -as block_imported

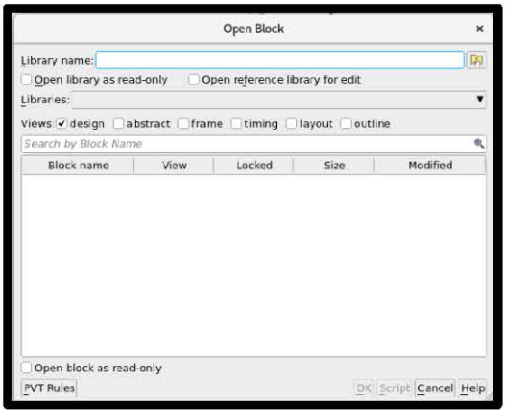
PHYSICAL DESIGN

FLOORPLAN

a. MANUAL

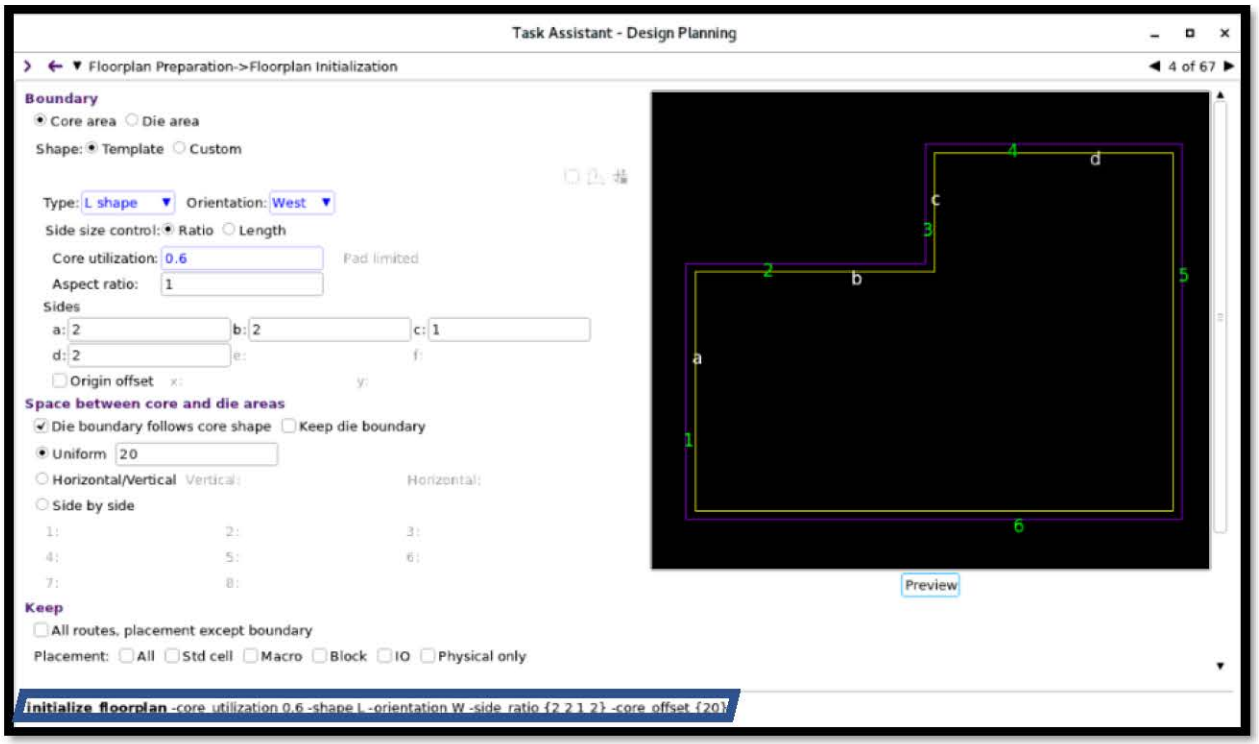
iv. To open the block through tool.





Note: Library Name to be the path of the project library and library name created in import design stage Designs will be listed and click on the block name block_imported and click on OK.

v. Creating Core Area and Die Area.





PHYSICAL DESIGN

vi. Placement of Ports

1. Setting the constraints for the ports

icc2_shell> set_block_pin_constraints -self -allowed_layers {M3 M4} -side { 2 } -hard_constraints location

2. Placing the ports

icc2_shell> place_pins -self

vii. Creating Voltage Area (For a Multi-Voltage Design)

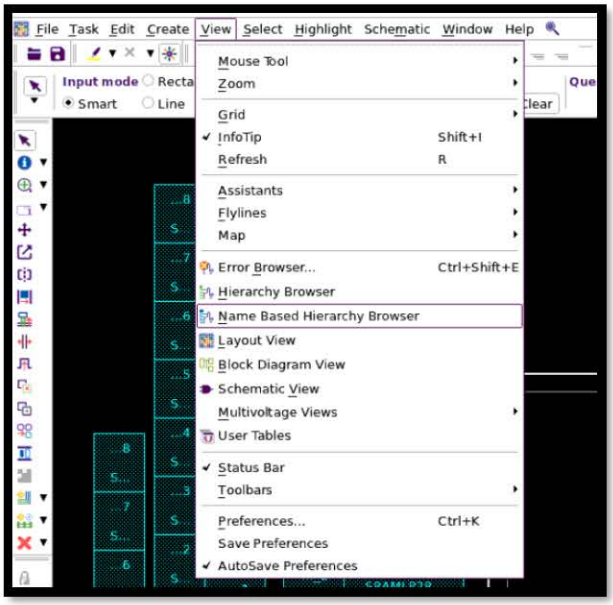
icc2_shell> create_voltage_area -power_domains PD_RISC_CORE -guard_band {{$a $b}} -region {{$llx $lly} {$urx $ury}}



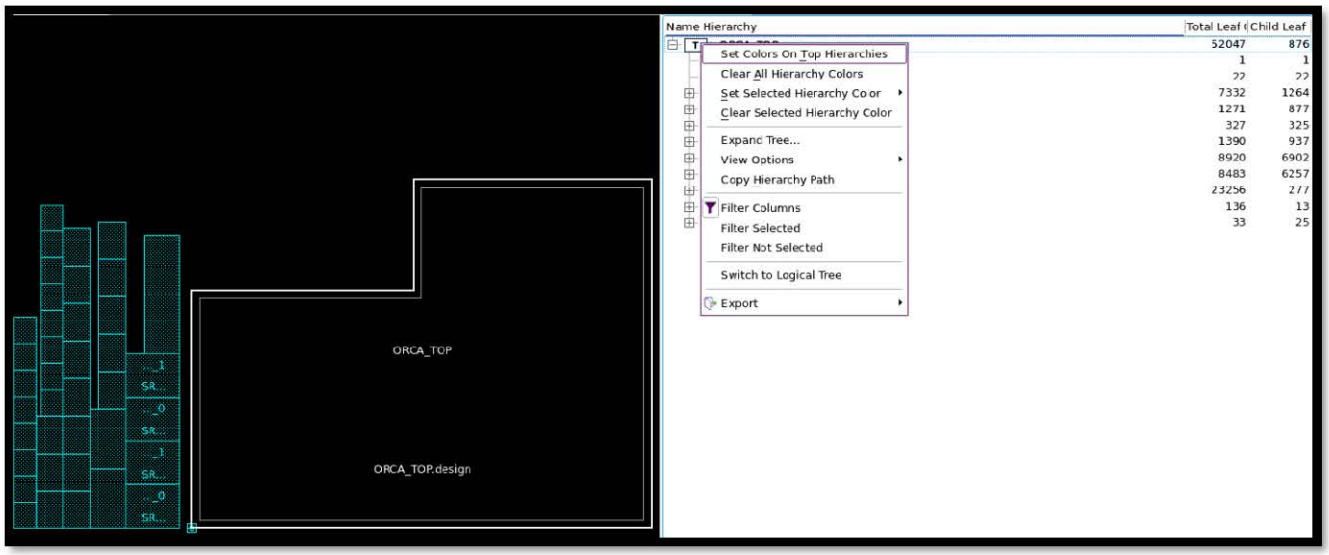
viii. Macro Placement

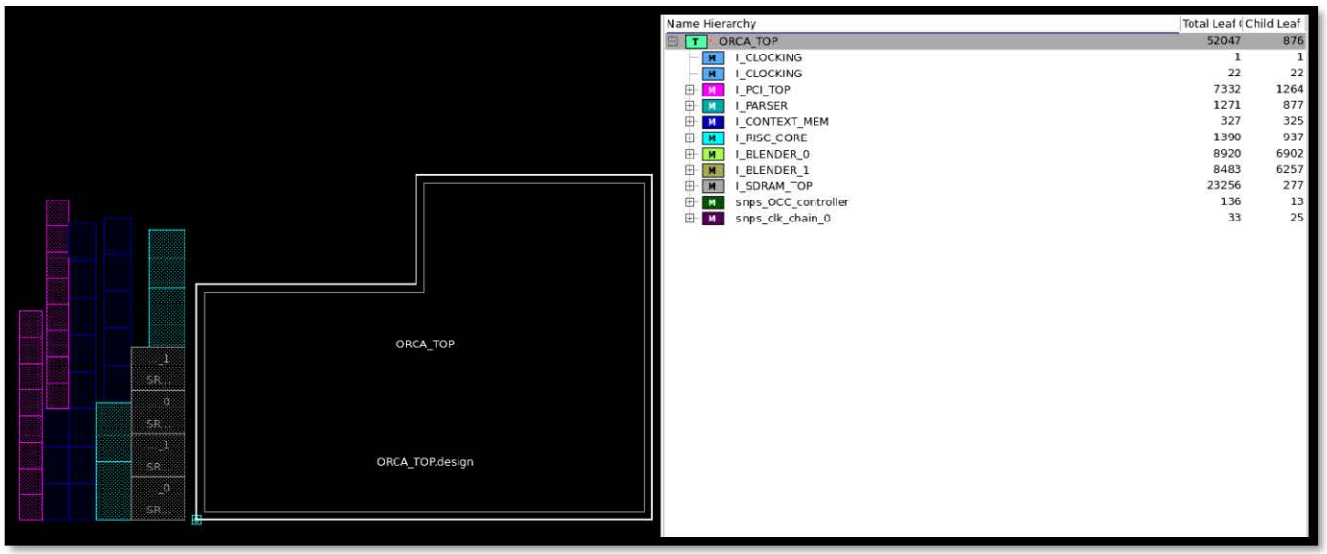
1. Enabling Different colours for different families

icc2_shell> set_colors -cycle_color

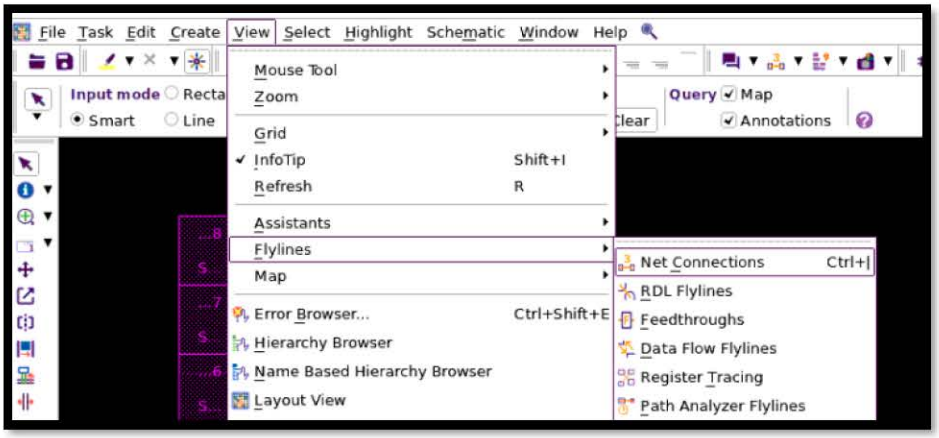


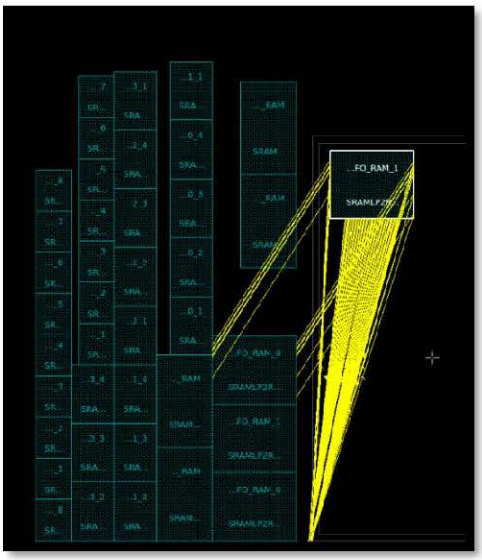
PHYSICAL DESIGN

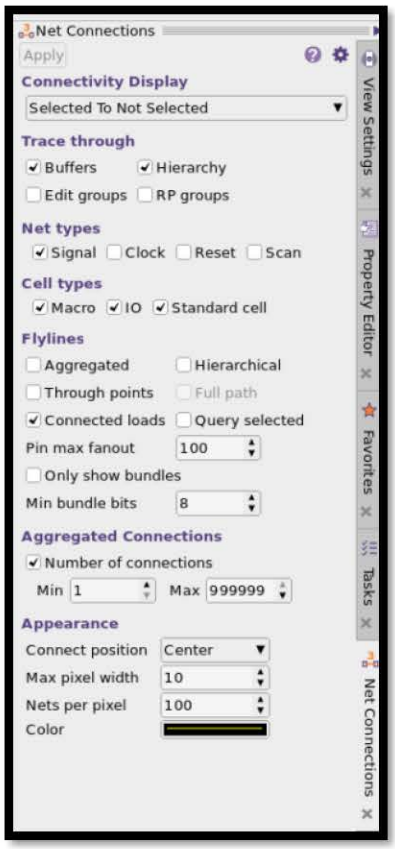




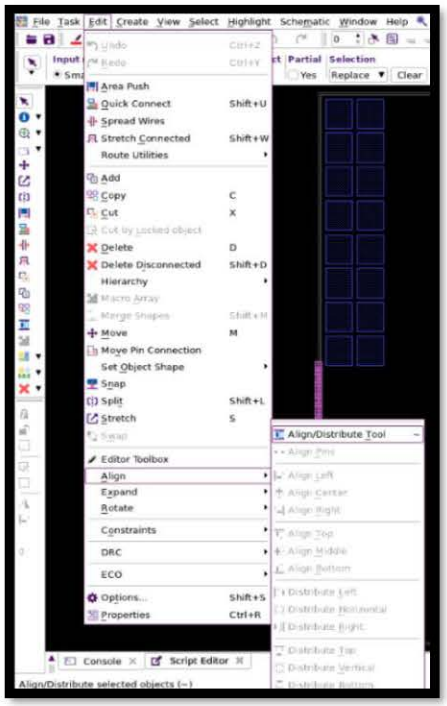
2. Enabling Fly Lines

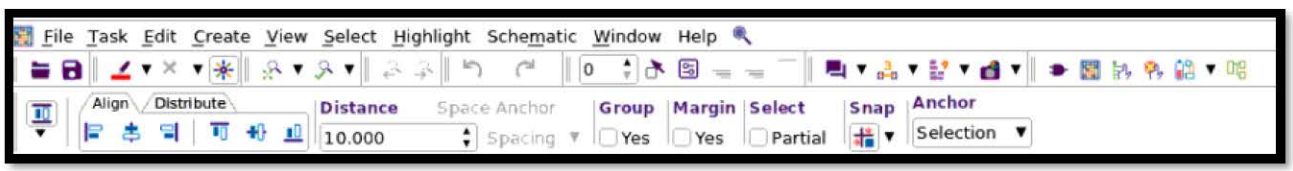






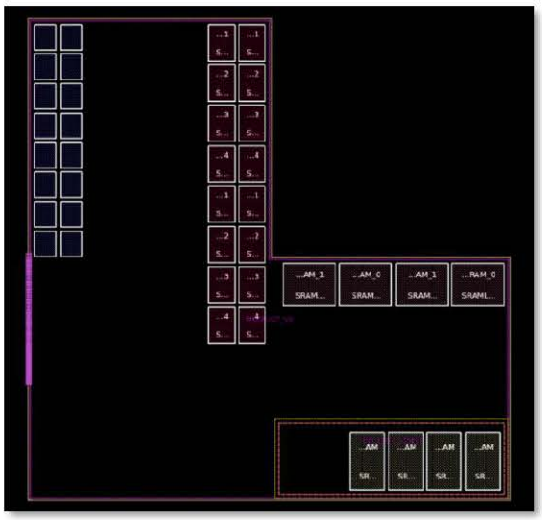
3. Align and Distribute tool





Note: Distance is the offset value.

4. Macro's Placed inside of core area



ix. Fixing of Macro's and Ports

#To Fix Macro's

icc2_shell> set_attribute [get_flat_cells -filter "is_hard_macro"] physical_status fixed

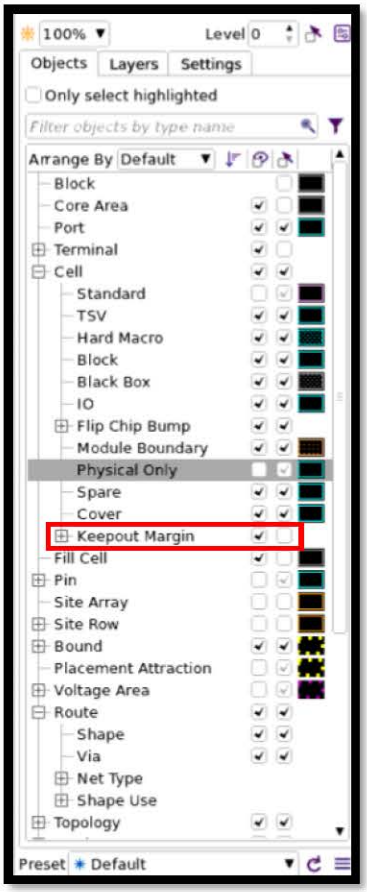
#To Fix Ports

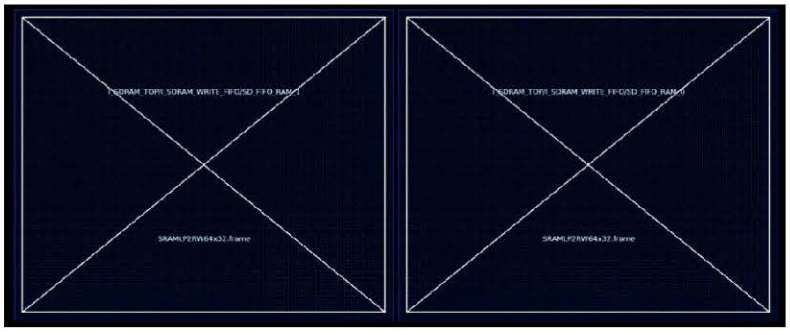
icc2_shell> set_attribute [get_ports *] physical_status fixed

x. Placing Blockages, Bounds and applying Keepout margin to the MACRO's

#To Apply KEEPOUT Margin to all the macro's in the design

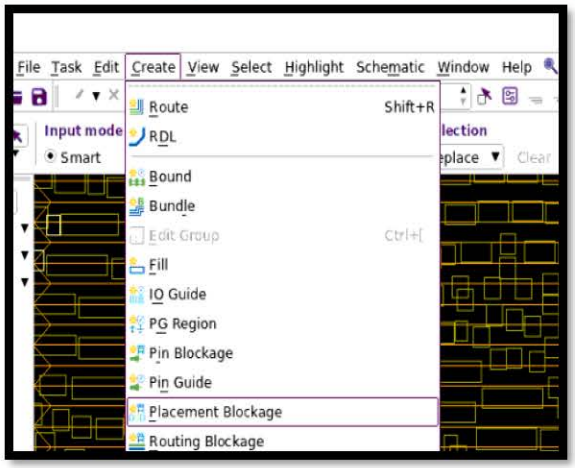
icc2_shell> create_keepout_margin -outer {3 3 3 3} [get_flat_cells -filter "is_hard_macro"]





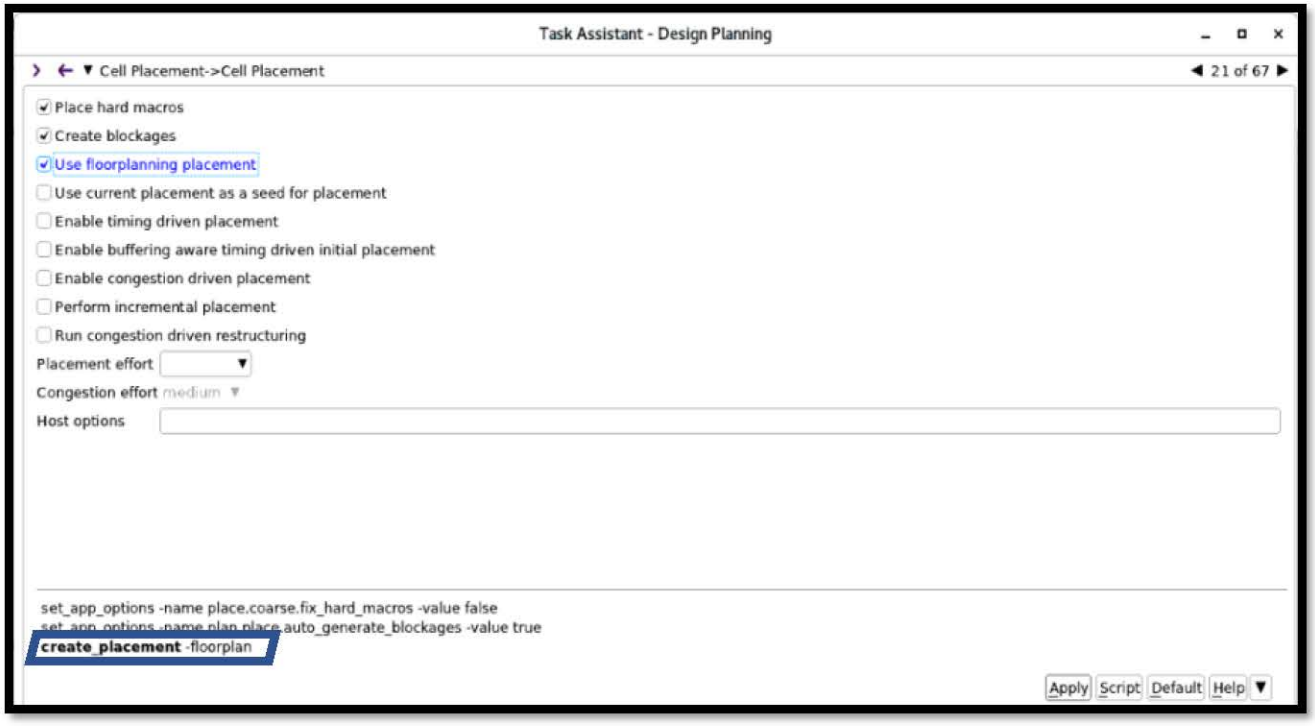
#To Create Placement Blockages

icc2_shell> create_placement_blockage -boundary {{$llx $lly} {$urx $ury}} -type hard





b. Automized Macro Placement



c. Save Design

d. Create DEF File

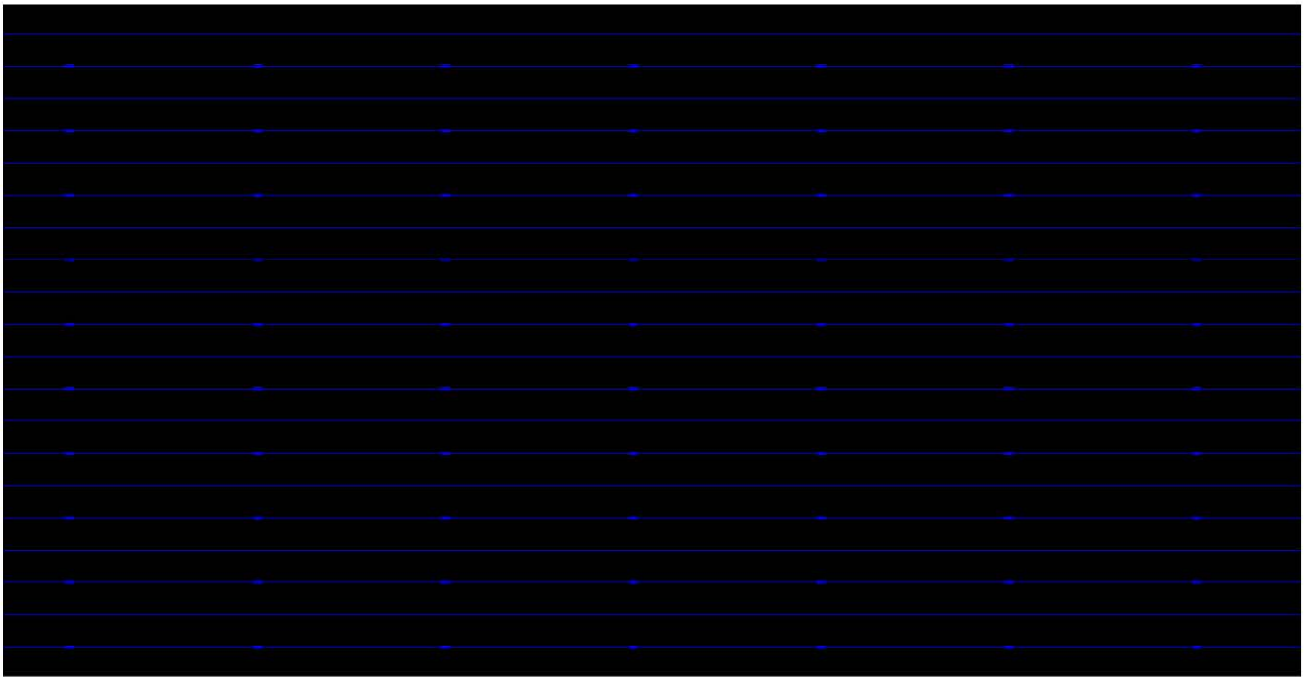
e. Creating of Rails

```
icc2_shell> create_pg_std_cell_conn_pattern P_std_cell_rail

icc2_shell> set_pg_strategy S_std_cell_rail_VSS_VDD -core -blockage {\

{{nets: VDD} {voltage_areas: PD_RISC_CORE}} {macros_with_keepout: $all_macros} } -pattern {{pattern:
P_std_cell_rail}{nets: {VSS VDD}}} -extension {{stop: outermost_ring}{direction: L B R T }}

icc2_shell> compile_pg -strategies {S_std_cell_rail_VSS_VDD}
```



f. To connect PG Pins of Macro's to the straps
g. Sanity Checks
   xi. Check whether stripes are formed all over the design
   xii. If any routing is done over macros? Check whether that routing is allowed or not? If it is not allowed remove the PG routing the macro's?
   xiii. Check for the floating wires and vias
   xiv. Check for DRC errors
h. Save Design
i. Create Def

```
icc2_shell> write_def ./outputs/power_plan.def
```

PHYSICAL DESIGN

PURPOSE	ICC2
To open block	a) open_lib lib_name b) open_block block_name
To get the design by reading the def file	a) open_lib lib_name b) open_block block_name c) read_def file_name.def
To create rings	a) create_pg_ring_pattern b) set_pg_stratergy c) compile_pg
To create stripes	a) create_pg_mesh_pattern b) set_pg_strategy c) compile_pg
To create rails	a) create_pg_std_cell_conn_pattern b) set_pg_strategy c) compile_pg
To save the design a) Library b) Block/cel c) Def file	 a) save_lib b) save_block -as powerplan_done c) write_def
Sanity checks	a) check_pg_missing_vias b) check_pg_drc -ignore_std_cells c) check_pg_connectivity -check_std_cell_pins none d) check_mv_design e) report_power

PHYSICAL DESIGN

PLACEMENT

a. Opening the design through reading the DEF file

icc2_shell> open_block ./outputs/work/orca_top.nlib:orca_top_imported
icc2_shell> read_def ./outputs/power_plan.def

b. Physical Cells

i. End Cap Cells

icc2_shell> create_boundary_cells -right_boundary_cell DCAP_HVT -left_boundary_cell DCAP_HVT

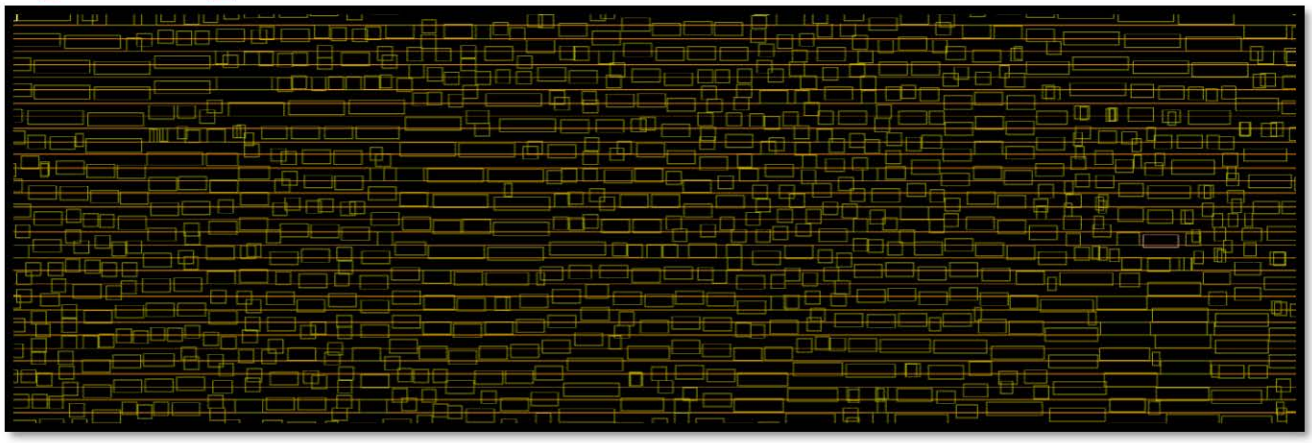
ii. Well-Tap Cells

icc2_shell> create_tap_cells -lib_cell SHFILL3_RVT.frame -distance 60 -pattern stagger -skip_fixed_cells

c. Types of placement

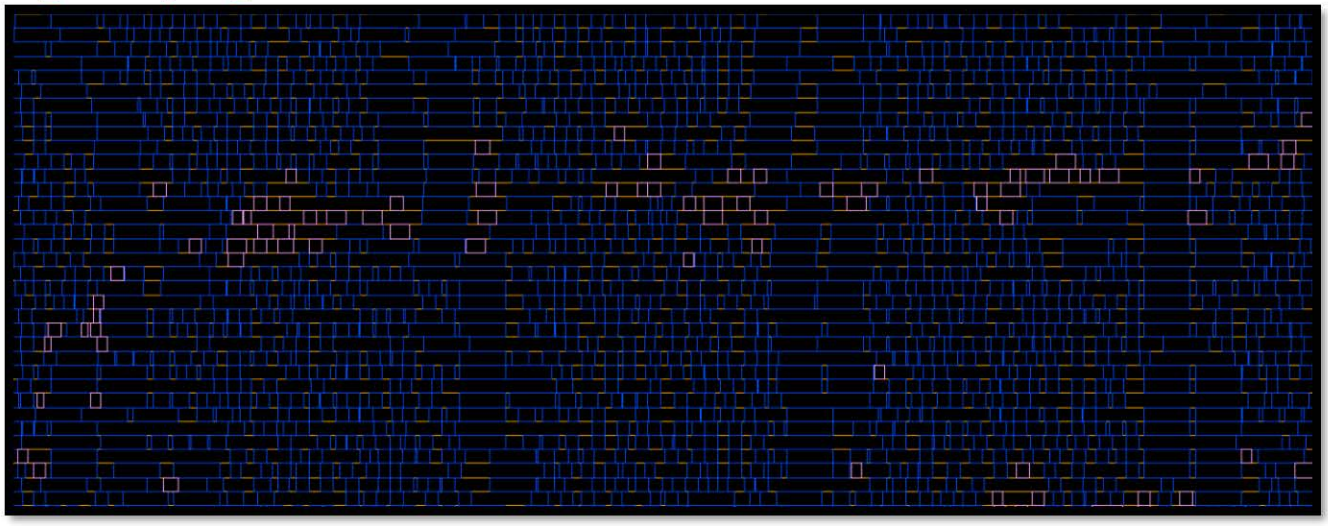
i. Coarse Placement: It will place the standard cells in the core area randomly

icc2_shell> create_placement -incremental



ii. Legalize Placement: It will place the standard cells in legalized location.

icc2_shell> legalize_placement



iii. Detail Placement

icc2_shell> place_opt -to final_opto

d. Save Design

iv. BLOCK

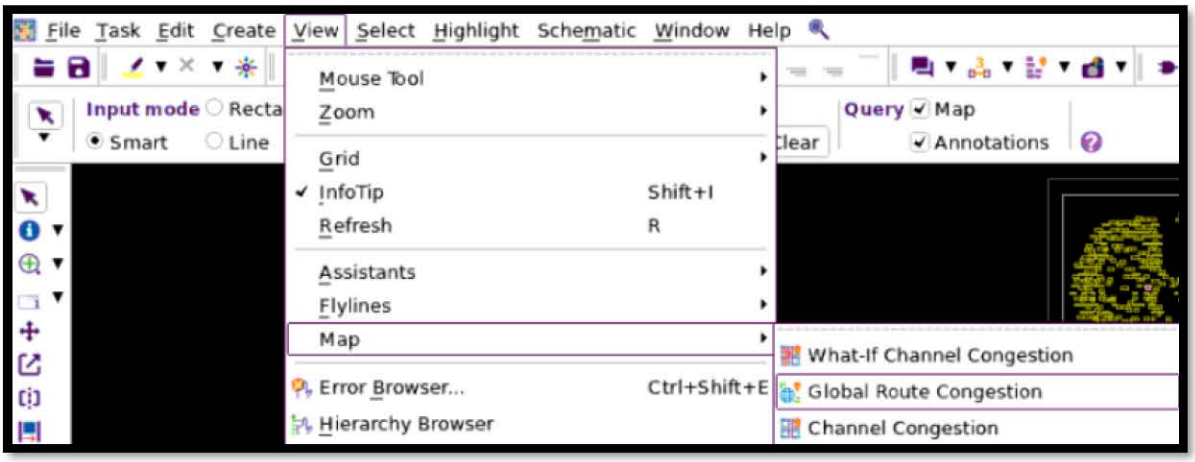
v. Def

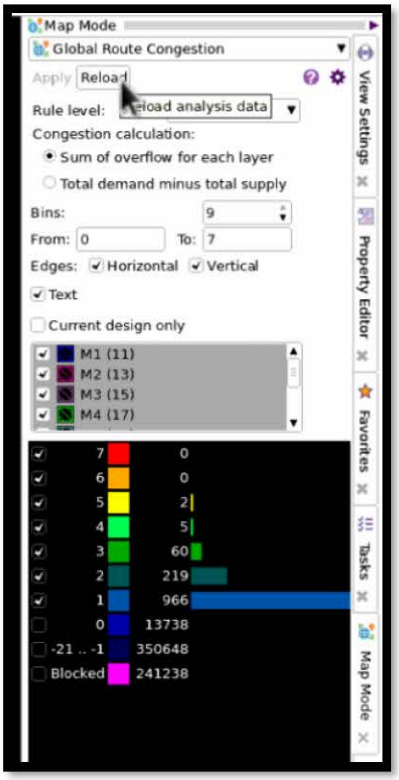
PHYSICAL DESIGN

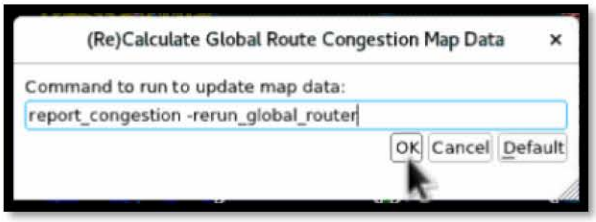
e. Sanity Checks

i. Congestion

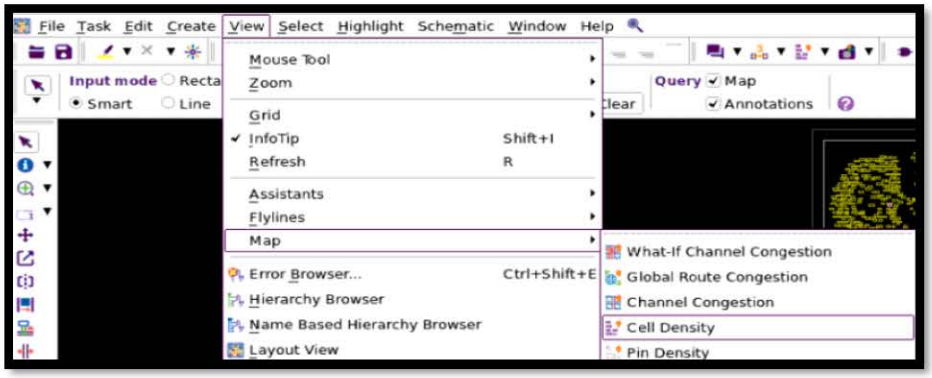
1. Global Routing Congestion



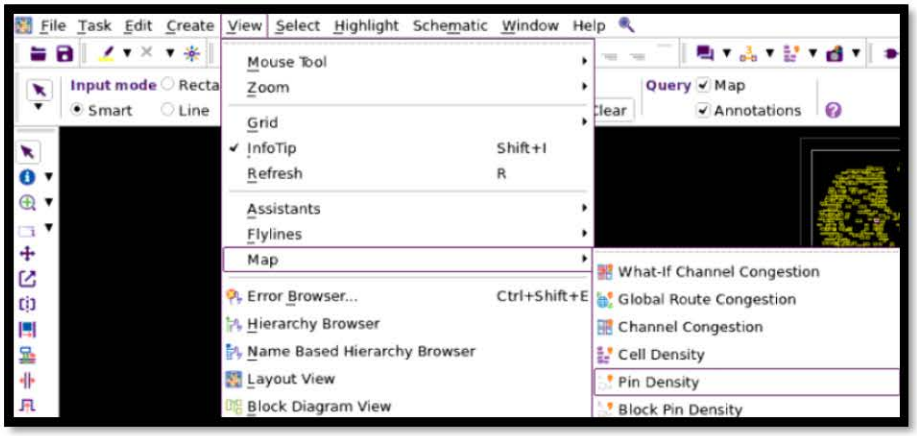




2. Cell Density



3. Pin Density



ii. Logical DRC
    1. Maximum transition
    2. Maximum Capacitance
    3. Maximum Fanout
iii. Timing Checks (Setup)

f. Fixes
    i. Swapping of cells (HVT to LVT/LVT to HVT)
    ii. Upsizing/ Downsizing of cells
    iii. Inserting Buffer
    iv. Cloning
    v. Path Group

PHYSICAL DESIGN

PURPOSE	ICC2
To OPEN BLOCK	open_lib lib_name open_block block_name
To get the design by READING THE DEF file	open_lib lib_name open_block block_name read_def file_name.def
COARSE PLACEMENT	create_placement -incremental
LEGALIZE PLACEMENT	legalize_placement
DETAIL PLACEMENT	place_opt
SANITY CHECKS a) GLOBAL ROUTING CONGESTION b) MAX TRANSITION c) MAX CAPACITANCE d) LEGALITY	 a) report_congestion -rerun_global_router b) report_constraints -max_transition c) report_constraints -max_capacitance d) check_legality
TIMING CHECKS	report_timing -delay_type max
To save the design a) Block b) Def file	 a) save_block -as place_done b) write_def

FIXES METHODS

PURPOSE	ICC2
To create BLOCKAGES	create_placement_blockage
To create KEEPOUT margin	create_keepout_margin
SWAPPING/SIZING of CELLS a) To get the alternative library cells available b) To get alternative drive strength with same flavour c) To get alternative VT cells with same drive strength d) To swap the cell	 a) get_lib_cells */ref_name* b) get_lib_cells */AND*_LVT c) get_lib_cells */AND2_*VT d) size_cell
To insert buffer	insert_buffer

PHYSICAL DESIGN

i. Sanity Checks
   xv. Before CTS
      1. Check whether height and width of block is defined
      2. Pins are placed and fixed
      3. Macros are placed and fixed
      4. Placement of standard cells is done
      5. Cells are placed legally
      6. Congestion should be very less
   xvi. After CTS
      1. Legality
      2. Logical DRC's
      3. Skew is balanced and insertion delay is less
      4. Timing Checks
         a. Setup
         b. Hold

PURPOSE	ICC2
To OPEN BLOCK	open_lib lib_name open_block block_name
To get the design by READING THE DEF file	open_lib lib_name open_block block_name read_def file_name.def
NDR RULES	create_routing_rule
To build CLOCK TREE and OPTIMIZE	clock_opt
To save the design a) Block/CEL b) DEF file	 a) save_block -as cts_done b) write_def
Sanity Checks Before CTS	a) check_design -checks pre_clock_tree_stage b) report_congestion c) report_timing d) report_constraints
Sanity Checks After CTS	a) check_clock_trees b) report_clocks -skew c) report_clock_qor d) check_legality e) report_congestion f) report_constraints g) report_timing -delay_type min h) report_timing -delay_type max

PHYSICAL DESIGN

PURPOSE	ICC2
Global Routing	route_global
Track Assignment	route_track
Detail Routing	route_detail
Automatic Routing based on Routing Steps	route_auto
To save the design a) Block/CEL b) DEF file	 a) save_block -as route_done b) write_def
Output Files a) Netlist b) SDC c) GDS-II	 a) write_verilog b) write_sdc c) write_gds
Sanity Checks	a) check_routes b) check_lvs c) report_qor -summary d) report_constraints e) report_timing -delay_type min f) report_timing -delay_type max

PHYSICAL DESIGN

REPORTS

1. Report Design
   icc2_shell> report_design -all

2. Report Units
   icc2_shell> report_units

3. Report Quality of Results
   icc2_shell> report_qor

4. Report Utilization
   icc2_shell> create_utilization_configuration core_util -scope block -include {all}
   icc2_shell> report_utilization -config core_util

5. Report Logical DRC's
   icc2_shell> report_constraints

6. To Report the Timing Slacks (Setup and Hold)
   icc2_shell> report_timing -delay_type max
   icc2_shell> report_timing -delay_type min

7. Power Report
   icc2_shell> report_power

8. Report Port
   icc2_shell> report_port

9. To Get Lib Cells
   icc2_shell> get_lib_cells

10. Legality
    icc2_shell> check_legality

11. Congestion Report
    icc2_shell> report_congestion

12. To report app options
    *icc2_shell> report_app_options **

13. To Report Voltage Areas
    icc2_shell> report_voltage_areas

14. To check PG connectivity
    icc2_shell> check_pg_connectivity

15. To get Report of Modes, Corners, PVT and Scenarios
    icc2_shell> report_modes
    icc2_shell> report_corners
    icc2_shell> report_scenarios
    icc2_shell> report_pvt